Linear Programming



LINEAR PROGRAM (SLIDE FOR AGENTIC CODING)

Minimize: $Z = 3x_1 + 4x_2$

Subject to:

$x_1 + 3x_2 >= 10$

$2x_1 + x_2 >= 6$

$x_1, x_2 >= 0$

Let’s try to solve a larger problem next.



DARK STORE LOCATION OPTIMISATION

- Entities
  - I: Set of customer (demand) locations, indexed by i
  - J: Set of candidate dark store locations, indexed by j

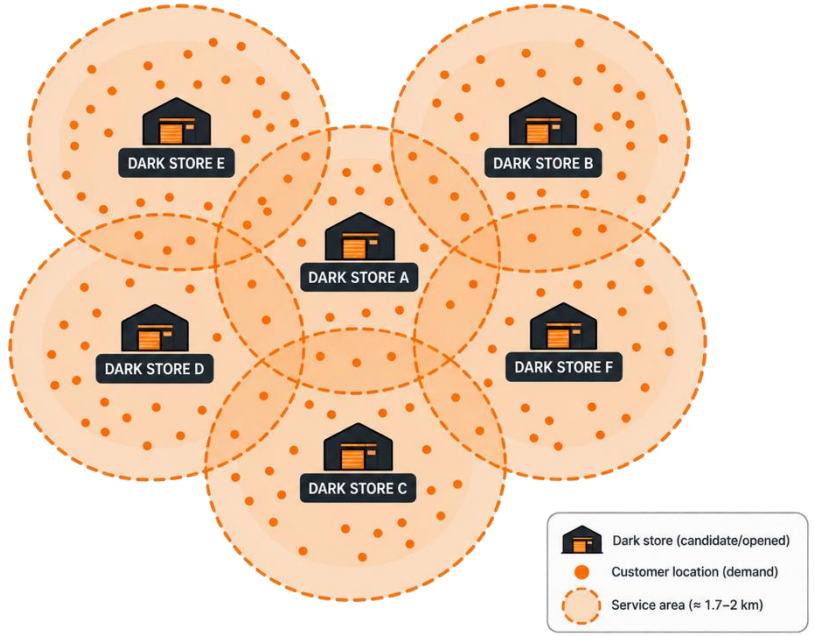
- Parameters/Inputs
  - $Demand_i$: Daily Demand (number of orders) at customer location i
  - $Capacity_j$: Daily Capacity (max orders per day) of dark store at location j
  - $FixCost_j$: Fixed cost of opening dark store a location j
  - $DelCost_{ij}$: Cost of delivering per unit of shipment from dark store j to customer i

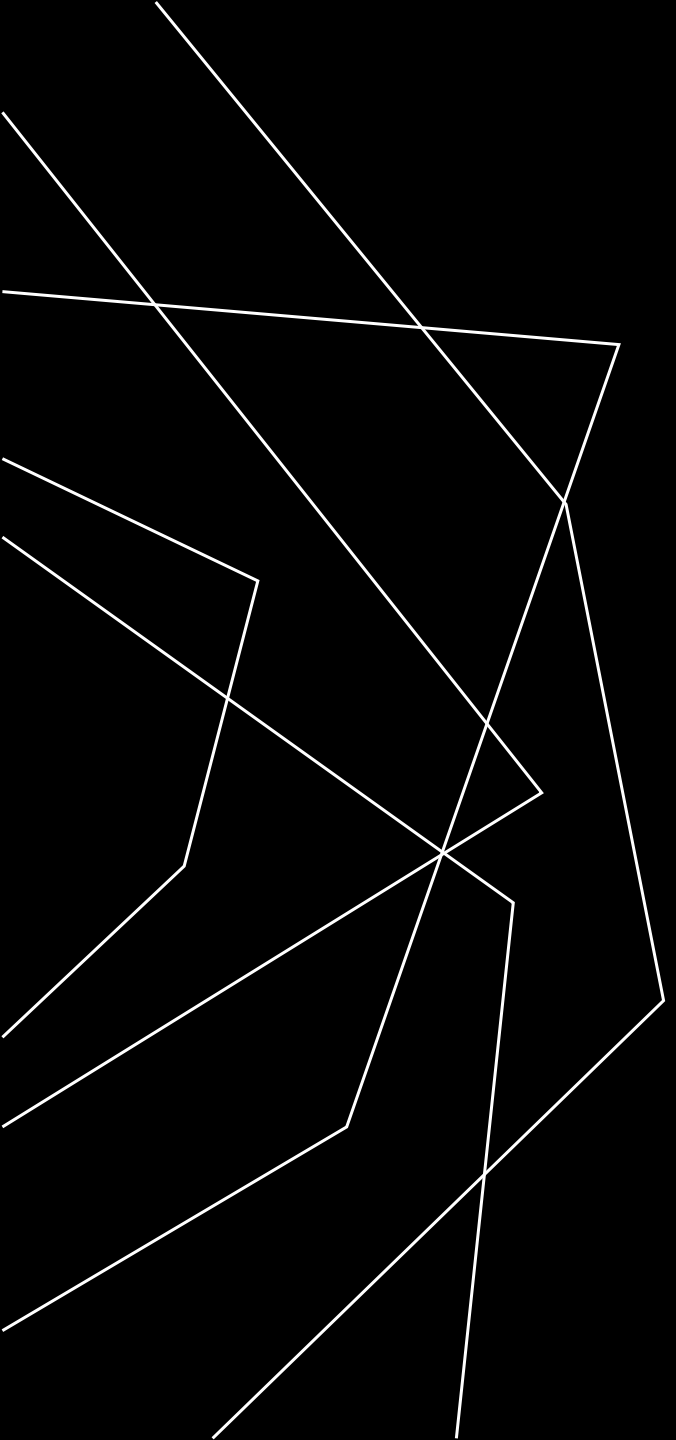
- Decision Variables:
  - $y_j \in \{0,1\}$: Binary variable to indicate if a dark store is opened at location j
  - $x_{ij} >= 0$. Share of daily demand of customer i served from dark store at location j



DARK STORE LOCATION OPTIMISATION

- Constraint 1: Demand Satisfaction
  - $\sum_{j \in J} x_{ij} = Demand_i, \quad \forall i \in I$
- Constraint 2: Facility Capacity
  - $\sum_{i \in I} x_{ij} <= Capacity_j \, y_j, \qquad \forall j \in J$
- Objective
  - Minimize $\sum_{i \in I} \sum_{j \in J} DelCost_{ij} x_{ij} + \sum_{j \in J} FixCost_j y_j$





THANK YOU

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